Finding report: Exact polynomial calculus degree p for the onto functional pigeonhole principle with surplus divisible by p

Lindheim, Björn

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1 Introduction

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Setting

The onto functional pigeonhole principle onto-FPHP$_n^{n+k}$ says that $n+k$ pigeons can be mapped one-to-one onto n holes. It is false for every $k \geq 1$. As a system of polynomial equations in Boolean variables x_{ij} (pigeon i sits in hole j) it has the axioms $\sum_j x_{ij} - 1$ (each pigeon sits in a hole), $\sum_i x_{ij} - 1$ (each hole holds exactly one pigeon), $x_{ij}x_{i'j}$ (no two pigeons share a hole) and $x_{ij}x_{ij'}$ (no pigeon sits in two holes). This is the linear encoding used in the literature on this family [1, 2]. These results are for the linear encoding only: translating the CNF clause by clause instead turns each pigeon clause into a polynomial of degree n [2], so degree bounds below n say nothing about that encoding.

A polynomial calculus (PC) refutation over the field $\mathbb{F}_p$ derives the polynomial 1 from the axioms and the Boolean axioms $x^2 - x$, using linear combinations and multiplication by a variable. Its degree is the largest degree of a line. The PC degree of the principle is the least degree of a refutation. When p does not divide k, summing the pigeon axioms and subtracting the hole axioms gives the constant $-k$, a degree-1 refutation. The question is what happens when p divides k.

Result

For every prime p, every c not divisible by p, and every $n \geq 2p-3$, the PC degree of onto-FPHP$_n^{n+pc}$ over $\mathbb{F}_p$ is exactly p: there is a refutation of degree p, and none of degree $p-1$. More generally, whenever p divides k and $n \geq 2p-3$, there is no refutation of degree $p-1$. The same formula therefore has PC degree p over $\mathbb{F}_p$ and PC degree 1 over $\mathbb{F}_q$ for every prime q not dividing pc.

Why it matters

Razborov's survey asks (Question 7) whether onto-FPHP$_n^{n+p^\ell}$ has bounded-degree PC refutations in characteristic p [2]. The upper bound p at the first level comes from Riis's criterion [3]. The known lower bounds are for the weaker Nullstellensatz system [1] and, for PC, a bound that grows with ℓ for large n [4]. This result fixes the degree exactly at the first level (surplus divisible by p but not by p^2), for every prime, from a small linear threshold in n.

Proved vs checked

Everything is proved in Lean 4 and kernel-checked, for all primes p and all $n \geq 2p - 3$; nothing is checked only up to a bound. The proofs use only the standard axioms `propext`, `Classical.choice` and `Quot.sound`. A second agent in the run re-proved each of the three statements in Lean, but its re-proofs reuse some of the author's lemmas (one to nine per proof, listed with each claim), so they are not fully independent. No person has reviewed the proofs; the evidence is the kernel check.

The Lean statements are for the library's formulation of PC, in which lines are functions on the Boolean cube. A later theorem of the same library shows that a refutation of degree D in standard (syntactic) PC over $\mathbb{F}_p$ exists exactly when one exists in the library's formulation, so the result holds verbatim for standard PC (`Programme.ń20261029-1156-proof-complexityż.StdPCExact.std_pc_degree_exact` in the published library; this finding's own checks do not re-check it).

Novelty statement

This result extends:

- Riis-Sitharam, ECCC TR97-048, Theorem 33: PC degree >= 2^l - 1 for onto-PHP^{N+p^l}_N over characteristic p once N >= (21+3p^2)(2^l - 2) - p (same family; proof not checked by us; at l = 2 this is >= 3 against our >= p+1 from n = 2p-1; at p = 2, l = 3 it is >= 7 for N >= 196 against our >= 5 for n >= 7).
- Beame-Riis 1998, Theorem 8.1 and Lemma 8.7: Nullstellensatz degree between 2^l - 1 and p^l - 1 (multiplier degree) for the same family; Section 10 lists polynomial-calculus lower bounds as open.
- Riis 1993, as Theorem 8 of Razborov's survey: degree-d PC refutations when C(m,d) and C(n,d) differ mod p, which gives the matching upper bound p for k = pc, p not dividing c (via Lucas).
- Razborov, Proof complexity of pigeonhole principles (2001), Question 7: the fixed-p, fixed-l bounded-degree question for onto-FPHP^{n+p^l}_n.
- Miksa-Nordstrom (JACM 2024): general PC degree method; onto-FPHP with m = n+1 easy (Riis); the case of characteristic dividing the surplus not covered.

- BGIP 2001: proves linear PC degree for Tseitin mod p over characteristic q containing a primitive root of unity of order p (Thm 15) and for MOD_p^n over F_q, q not dividing p (Cor 20: "requires degree > delta n"). This is the cross-characteristic regime. Our results concern the same-characteristic regime p | k, so there is no overlap. The paper does not treat onto-PHP.

- Alekhnovich-Razborov 2003: "immunity" method giving linear PC degree bounds for mod_p Tseitin over characteristic different from p, for flow tautologies and for random k-CNF in characteristic 2 (abstract only). No PHP or onto-PHP result. The method is relevant only as possible technique.

- Impagliazzo-Pudlak-Sgall 1999: simpler proof of Razborov's ceil(n/2)+1 PC lower bound for (injective, non-onto) PHP^m_n over any field, plus an upper bound in large characteristic (abstract). This is a different principle: without the onto and functional axioms the bound is independent of p. It does not transfer to the exact degree p or p+1 for onto-FPHP.

- BIKPRS 1996/97: Nullstellensatz lower bounds for Count_q over Z_m with q not dividing m (Thm 3.2) and over Q (Thm 4.6), plus the PC-to-NS simulation (Thm 5.4). This is Nullstellensatz and Count, not PC degree for onto-FPHP, and the paper excludes the case q | m.

- Grigoriev FOCS 1998 (snippet only, not opened): Nullstellensatz degree for Tseitin mod p over other characteristics. It is a predecessor of BGIP; NS, not PC, and a different principle.

- Filmus-Lauria-Mika-Nordström-Vinyals 2025: PC space and separations. It mentions the functional PHP only for space limits, not for characteristic-dependent degree.

- Ken 2022/2024: restates Razborov's injPHP bound "no PC-refutation ... with degree <= m/2" (Thm 24) and extends it to NS over finite rings (Cor 25). Onto-PHP appears only in reductions (Props 12, 15), and no degree bound for it is proved.

- Reduction transfer: none of the opened sources gives a low-degree reduction from Count_q, Tseitin mod q or Mod_q (q != p) to onto-FPHP^{n+k}_n with p | k over F_p. Since the cross-characteristic lower bounds need q != p, they do not yield results 1-3 directly.

Our prior-art search (2026-11-06) covered:

- Riis, Sitharam, Non-constant degree lower bounds imply linear degree lower bounds, ECCC TR97-048 (1997), pp. 36-38 and 45-46 (Theorem 33, Propositions 27 and 28), opened 2026-10-29

- Beame, Riis, More on the relative strength of counting principles, DIMACS Series 39 (1998), homes.cs.washington.edu/~beame/papers/riis.pdf: Theorem 8.1, Lemma 8.7, Section 10, opened 2026-10-27 and 2026-11-01
- Razborov, Proof complexity of pigeonhole principles, DLT 2001 (LNCS 2295), people.cs.uchicago.edu/~razborov/files/php_survey.pdf: Theorem 8, Question 7, footnote 5, opened 2026-10-27 and 2026-10-29
- Riis, BRICS Report RS-99-29, opened 2026-10-29
- Miksa, Nordstrom, A generalized method for proving polynomial calculus degree lower bounds (CCC 2015; JACM 2024; arXiv 1505.01358), opened 2026-10-27
- de Rezende et al., Theory of Computing 21(6) (2025), opened 2026-10-29
- Lauria et al., Theory of Computing 21(4) (2025), opened 2026-10-27
- Buss, Grigoriev, Impagliazzo, Pitassi, "Linear gaps between degrees for the polynomial calculus modulo distinct primes", JCSS 62(2):267-289, 2001 (STOC 1999). Opened: journal final PDF, https://mathweb.ucsd.edu/~sbuss/ResearchWeb/lineargaps/j
- Alekhnovich, Razborov, "Lower bounds for polynomial calculus: non-binomial case", Proc. Steklov Inst. Math. 242 (2003) 18-35 (FOCS 2001). Opened: abstract only, https://mathnet.ru/eng/tm403 (full text not opened).
- Impagliazzo, Pudlak, Sgall, "Lower bounds for the polynomial calculus and the Groebner basis algorithm", Computational Complexity 8(2):127-144, 1999 (ECCC TR97-042). Opened: ECCC abstract only, https://eccc.weizmann.ac.il/report/1997/04
- Buss, Impagliazzo, Krajicek, Pudlak, Razborov, Sgall, "Proof complexity in algebraic systems and bounded depth Frege systems with modular counting", Computational Complexity 6(3):256-298, 1996/97. Opened: full PDF, https://mathweb.ucsd.edu/~sbuss/ResearchWeb/nullstellensatz/token.pdf
- Grigoriev, "Tseitin's tautologies and lower bounds for Nullstellensatz proofs", FOCS 1998, pp. 648-652. Snippet only, not opened (dblp/researchr entries).
- Mika, Nordström, "A generalized method for proving polynomial calculus degree lower bounds", CCC 2015 (arXiv 1505.01358). Already credited. Seen in search results; not re-opened here.
- Filmus, Lauria, Mika, Nordström, Vinyals, "Towards an understanding of polynomial calculus: new separations and lower bounds", Theory of Computing 21(4), 2025 (ICALP 2013). Opened: abstract page, https://zesty2.cs.uchicago.edu/articles/v021a0

- Eitetsu Ken, "On some Sigma_0^B-formulae generalizing counting principles over V^0", arXiv 2203.10237 (2022; a journal version appears to be Arch. Math. Logic 2024, https://link.springer.com/article/10.1007/s00153-024-00938-1). Opened: ar5iv HTML and Springer page, first ~100k characters of each.
- Web searches for 2000-2026 papers on PC degree of onto-PHP or Count where the characteristic divides the surplus: no further hits beyond the items above.

Prior-art answer recorded by the owner (basis stated in the file); published in `prior-art/`.

If you know of related work we missed, please open an issue.

This result was found by AI research agents (the autoresearch harness). It was proved in Lean; the verification steps below build the published programme library.

How it was produced

The declaring Lean sources are in `checks/library/`.

How to verify

Lean toolchain: `leanprover/lean4:v4.35.0`. `https://github.com/kodakodapa/autoresearch-findings/tree/ad1259e5894f224a3d1eda62a1914da4b4dd699e/programmes/proof-complexity`. Install elan and open a new shell after installing elan, then pin the library subtree while keeping the finding checks:

```
git clone https://github.com/kodakodapa/autoresearch-findings.git
cd autoresearch-findings
git checkout ad1259e5894f224a3d1eda62a1914da4b4dd699e -- programmes/proof-complexity
cd programmes/proof-complexity && lake build 'Programme.ń20261027-0607-proof-complexityż.Luc
AR_LIBRARY=$PWD python -I ../../claims/exact-pc-degree-onto-fphp-first-level/checks/lean-K13
AR_LIBRARY=$PWD python -I ../../claims/exact-pc-degree-onto-fphp-first-level/checks/lean-K14
AR_LIBRARY=$PWD python -I ../../claims/exact-pc-degree-onto-fphp-first-level/checks/lean-K15
```

A full `lake build` builds the whole library. The named targets build only the claim modules and their dependencies. Indicative measurements from the 2026-11-06 reproduction on 2 cores / 7.8 GB: K13–K15 90 s; K1–K6 440 s; K18 87 min (90 certificate modules). Peak memory was 4.7 GB for one Lean process. Build times depend on hardware and cached dependencies.

`checks/run-copies/`: the author's working copies from the run; they import the run's private `Work` package and are not built standalone.

2 Claims

Claim K13

`Work.Lucas.pc_degree_exact`: for every prime p (`IsPrime p`), every c with $p \nmid c$, every $n \geq 2p-3$ and every D: `ontoFPHP n (p*c)` has a degree-D `PPCRefutation p` iff $p \leq D$ (PC degree over $\mathbb{Z}/p$ of onto-FPHP$_n^{n+pc}$ is exactly p).

Status: proved (axioms: propext, Classical.choice, Quot.sound) ů second-agent re-proof (skeptic/codex).

Re-checked at publication on 2026-11-08: pass (`results/publication/lean-K13.json`).

Verified. No check scripts.

Evidence.

- library check `checks/lean-K13.py` (library theorem `Programme.ń20261027-0607-proof-complexityż.I`
- proof `checks/run-copies/K13-Lucas.lean`: Work.Lucas.pc_degree_exact — pass (`results/K13-2.json`)
- reasoning: Lower bound: K14 (no degree $p-1$) plus monotonicity of PPCDerivable in the degree. Upper bound: library OntoLucas.ontoFPHP_upper at $t = p$, whose unit hypothesis is $\binom{n+pc}{p} - \binom{n}{p} \equiv c \not\equiv 0 \bmod p$; proved by $p \mid \binom{p}{r}$ for $0 < r < p$, Pascal's rule and telescoping in c. Convention: PC line degree, functional encoding. This agrees with Beame-Riis Lemma 8.7: their NS degree $p^l - 1$ counts multiplier degree, and the products have degree p^l, which is line degree p at $l = 1$.
- reproof `checks/reproofs/K13-ExactReproof.lean` by the skeptic (codex): `Work.Lucas.pc_degree_exact` — pass (`checks/reproofs/K13-ExactReproof.json`); second-agent re-proof; uses 2 lemmas from the author's package; re-proof by a second agent; imports the run's private packages (not rebuildable from this repo)

Prior art. Partial prior art.

Claim K14

`Work.Peel.pc_degree_ge_p`: for every prime p (`IsPrime p`), every k with $p \mid k$ and every $n \geq 2p-3$, `ontoFPHP n k` (onto-FPHP$_n^{n+k}$, functional encoding) has no `PPCRefutation p` of degree $p-1$ (polynomial calculus over $\mathbb{Z}/p$).

Status: proved (axioms: propext, Classical.choice, Quot.sound) ů second-agent re-proof (skeptic/codex).

Re-checked at publication on 2026-11-08: pass (`results/publication/lean-K14.json`).

Verified. No check scripts.

Evidence.

- library check `checks/lean-K14.py` (library theorem `Programme.ń20261027-0607-proof-complexityż.I`
- proof `checks/run-copies/K14-Peel.lean`: Work.Peel.pc_degree_ge_p — pass (`results/K14-2.json`)
- reasoning: Paper proof (mirrors the Lean chain). Library criterion K4: if $\mathrm{EXT}_p(n_0, k, d)$ holds for all $d \leq p-2$ then no degree-$(p-1)$ refutation for any $n \geq n_0$. EXT at level d is proved by peeling: the flag lemma (K8) turns a balanced f into a tower of links; at level 0 subtract a symmetric window, and at level j subtract a block-product window (j rectangles $\times$ unit blocks $\times$ a symmetric degree-$(d-j+1)$ design on a rest block), which equals $(d-j+1)! \cdot RP$ on j-matchings. The residual has zero links, so S_j (K10) puts it in the span of rectangle products. Every factorial used is at most $(p-1)!$, a unit mod prime p; the blocks fit when $2d+1 \leq n$, i.e. $n \geq 2p-3$. ExtLin.EXT_of_span (K5) closes the step.
- reproof `checks/reproofs/K14-LowerReproof.lean` by the skeptic (codex): `Work.Peel.pc_degree_ge_p` — pass (`checks/reproofs/K14-LowerReproof.json`); second-agent re-proof; uses 9 lemmas from the author's package; re-proof by a second agent; imports the run's private packages (not rebuildable from this repo)

Prior art. Partial prior art.

Claim K15

`Work.Lucas.separation`: for primes p, q, $p \nmid c$, $q \nmid pc$, $n \geq 2p-3$: `ontoFPHP n (p*c)` has a degree-D `PPCRefutation p` iff $p \leq D$, and has a degree-1 `PPCRefutation q`.

Status: proved (axioms: propext, Classical.choice, Quot.sound) ů second-agent re-proof (skeptic/codex).

Re-checked at publication on 2026-11-08: pass (`results/publication/lean-K15.json`).

Verified. No check scripts.

Evidence.

- library check `checks/lean-K15.py` (library theorem `Programme.ń20261027-0607-proof-complexityż.I`
- proof `checks/run-copies/K13-Lucas.lean`: Work.Lucas.separation — pass (`results/K15-2.json`)
- reasoning: Over $\mathbb{F}_p$: K13. Over $\mathbb{F}_q$ with $q \nmid pc$: library OntoFPHP.ontoFPHP_deg1 needs a unit k^{-1} mod q, supplied by Bezout. The gap is p versus 1 and is constant in n for fixed p, so this separates characteristics along the parameter p, not asymptotically in n.

- reproof `checks/reproofs/K15-SeparationReproof.lean` by the skeptic (codex): `Work.Lucas.separation` — pass (`checks/reproofs/K15-SeparationReproof.json`); second-agent re-proof; uses 1 lemmas from the author's package; re-proof by a second agent; imports the run's private packages (not rebuildable from this repo)

Prior art. Partial prior art.

Provenance. Run `20261027-0607`, claims K13, K14, K15, run-folder commit `908b30496168c757d5eba6ba4ba761dbab7933b8`, published 2026-11-06.

3 Discussion

The degree here is a constant, p. The result says nothing about growth with ℓ, which is what Question 7 is about. Riis and Sitharam's Theorem 33 gives degree at least $2^\ell - 1$ for surplus p^ℓ once $N \geq (21 + 3p^2)(2^\ell - 2) - p$ [4]. At $\ell = 1$ that bound is 1, and surplus pc with $c > 1$ is outside its family. We have not checked the proof of Theorem 33. Below $n = 2p - 3$ the degree is not determined here. The Nullstellensatz degree of Beame and Riis [1] is multiplier degree, one less than the line degree used here.

4 How to verify

For programme findings, follow the clone, build and run steps in each finding. The declaring Lean sources are in `checks/library/`; the publication checks import their modules and print each theorem's axioms. The output of the checks re-run at publication is in `claims/<slug>/results/publication/`. The checks are Python scripts and need the packages of the research image (the image's `requirements.txt`); the Lean checks also need `lean`.

5 Revisions

- 2026-11-06 — Math rendering on the site; paper lists only cited references
- 2026-11-06 — Library published buildable; verification steps; run copies labelled
- 2026-11-06 — Verification steps; stale run copies removed; encoding stated
- 2026-11-06 — Publication re-checks and reproofs published; prior art extended; texts corrected
- 2026-11-06 — Library theorem names and checked statements shown; reproof wording corrected; build cost
- 2026-11-08 — Prior-art wording; library names; monotonicity

References

[1] Paul Beame and Søren Riis. More on the relative strength of counting principles. In *Proof Complexity and Feasible Arithmetics*, volume 39 of *DIMACS Series in Discrete Mathematics and Theoretical Computer Science*, pages 13–35. American Mathematical Society, 1998.

[2] Alexander A. Razborov. Proof complexity of pigeonhole principles. In *Developments in Language Theory (DLT 2001)*, volume 2295 of *Lecture Notes in Computer Science*, pages 100–116. Springer, 2002.

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[4] Søren Riis and Meera Sitharam. Non-constant degree lower bounds imply linear degree lower bounds. Technical Report TR97-048, Electronic Colloquium on Computational Complexity, 1997.